

SARTA handles **cloud scattering** through a highly optimized operational technique known as the **two-stream approximation**. While full multiple-scattering calculations (like Discrete Ordinates or Monte Carlo) are too slow for real-time satellite processing, SARTA simplifies the geometry to keep calculations lightning-fast. [1, 2, 3, 4, 5]

SARTA treats cloud layers by parametrizing their optical properties and inserting them into the fast radiative transfer framework through specific steps: [6]

1. The Multi-Stream to Two-Stream Simplification

- **Directional Fluxes:** SARTA compresses complex, multi-directional light scattering down to two principal paths: upward-welling and downward-welling radiation streams. [1, 3]
- **The Delta-Eddington Scaling:** It often incorporates delta-scaling to account for the strong forward-scattering peak typical of large cloud droplets and ice crystals. This transforms highly asymmetrical scattering into a much simpler mathematical problem. [7, 8, 9, 10]

2. Cloud Representation Predictors

To execute this scattering algorithm, SARTA expands its input requirements to include a specialized set of cloud parameter variables for up to two distinct cloud layers (typically a liquid water cloud and an ice cloud): [11, 12, 13]

- **Cloud Top Pressure (CTP):** Determines where in the atmospheric profile the scattering layer begins.
- **Cloud Mean Effective Particle Size (\$D_e\$ or \$r_e\$):** The average radius or diameter of the ice crystals or liquid water droplets. This dictates the scattering efficiency and phase function. [4, 7, 14, 15, 16]
- **Cloud Optical Depth (τ):** Quantifies the thickness of the cloud layer, determining how much radiation is absorbed versus scattered. [4, 17, 18, 19]
- **Cloud Fraction (\$C_f\$):** The percentage of the satellite pixel covered by the cloud, used to compute a weighted average of clear and cloudy radiances. [20, 21, 22]

3. Lookup Tables (LUTs) for Fast Execution

- SARTA cannot compute Mie scattering equations for water droplets or complex geometric phase functions for ice crystals on the fly.
- Instead, it uses pre-calculated **Lookup Tables** mapped across wavelengths, particle sizes, and ice/liquid phases.
- The model matches the input cloud properties to the table to instantly extract the necessary single-scattering albedo and asymmetry parameters. [7, 23, 24, 25, 26]

Limitations

While the SARTA two-stream scattering parameterization works remarkably well for standard liquid and ice clouds across the longwave infrared spectrum, it struggles with **Deep Convective Clouds (DCCs)**. Because DCCs are optically thick and vertically massive, the simplified

two-stream geometry breaks down, resulting in larger brightness temperature errors. [2, 27, 28]

Would you like to examine how SARTA combines these cloudy streams with **gaseous transmittances**, or look into its **shortwave infrared** capabilities? [2, 20]

- [1] <https://repository.library.noaa.gov>
- [2] <https://airs.jpl.nasa.gov>
- [3] <https://www.nature.com>
- [4] <https://www.nature.com>
- [5] <https://journals.ametsoc.org>
- [6] <https://agupubs.onlinelibrary.wiley.com>
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